BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Note: Objective Type

Q1. The anhydride of CTO3 is:

(A) CHO
(B) Cl2Or
(C) CO
(D) C

Q2. Benzene can not undergo

(A) Substitution reaction
(B) Addition reaction
(C) Oxidation reaction
(D) Elimination reaction

Q3. Which of these polymers is an addition polymer

(A) Nylon-6 6
(B) Epoxy resin
(C) Polystyrene
(D) Terylene

Q4. Phenol-formaldehyde resin is called:

(A) Bakelite
(B) Tellen
(C) Orion
(D) Terylene

Q5. The oxides of metals are:

(A) Acidic
(B) Amphoteric
(C) Neutral
(D) [illegible in source scan]

Q6. The carbon atom of a carboxyl group is:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q7. Tincal is a mineral of:

(A) Al
(B) B
(C) Si
(D) C

Q8. According to Lewis concept ethers behave as

(A) Acid
(B) Acid as well as base
(C) Base
(D) None of these

Q9. The residence time of NO in atmosphere is:

(A) One day
(B) Two days
(C) Three days
(D) Four days

Q10. A double bond consists of:

(A) Two sigma bonds
(B) One sigma and two pi bonds
(C) One sigma and one pi bond
(D) Two pi bonds

Q11. Which of the following sulphate is not soluble in water

(A) Sodium sulphate
(B) Potassium sulphate
(C) Zinc sulphate
(D) Barium sulphate

Q12. Vinyl acetylene combines with ICT to form

(A) Polyacetylene
(B) Benzene
(C) Chloroprene
(D) Divinyl acetylene

Q13. The state of hybridization of carbon atom in methane is

(A) sp
(B) sp
(C) ap
(D) dsp

Q14. Which one of the following is not nucleophile:

(A) H2S
(B) HO
(C) BF
(D) NI

Q15. Laughing gas is chemically:

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q16. Which test is applied to detect amino acids

(A) Fehling's test
(B) [illegible in source scan]
(C) Ninhydrin test
(D) Sodium nitroprusside test

Q17. Which is not a calcarious material

(A) Lime
(B) Chy
(C) Marble
(D) Marine shell

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Write down the chemistry of borax bead test.

(ii) Draw the electronic structure of CO and CO

(iii) How will you convert boric acid into borax?

(iv) Convert benzene into glyoxal

(v) Mention the product when phenol is distilled with Zn dust by giving reaction

(vi) Give two uses of silicones.

(vii) Define saponification number.

(viii) How polyvinyl acetate is formed? Write its equation.

(ix) Draw the structure of cholesterol.

(x) Write down the equation, when suspended impurities are removed in the colloidal form in raw water

(xi) Mention the hazards of chloroform

(xii) Define acid rain.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) Give the two reactions for the preparation of NO.

(ii) Write down four uses of HlNO3.

(iii) Give the reactions of H2SO4 with: (a) NaCl (b) KNO3

(iv) Convert CH4 into formaldehyde by catalytic oxidation.

(v) Define Markownikov's rule. Give an example

(vi) Prepare conide from ethene.

(vii) Name two main factors which govern reactivity of R-X bond in alkyl halides

(viii) Define nucleophile. Give two examples.

(ix) What is vital force they?

(x) What are heterocyclic organic compounds? Give two examples.

(xi) Write down four essential qualities of a good fertilizer.

(xii) Write down two chemical reactions involved in the preparation of urea.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Complete and balance the following chemical equation: KMnO4 + FeSO2+H2SO1 ->

(ii) Give systematic names of the following (a) KCw(CN) (b) K2[PlC]

(iii) What is meant by "central metal ion"? Explain with one example

(iv) How are ethene and diethyl ether produced from ethyl alcohol?

(v) Explain Lucas test.

(vi) Give reactions of phenol with: (a) Bromine water.

(vii) What is Benedict's solution test? Give reaction.

(viii) How does hydrazine react with acetone?

(ix) Write reaction between acetic acid and ammonia for the formation of amide

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [4] Define hydration energy. Give its trend in the periodic table.

(b) [4] Explain peculiar behaviour of Beryllium.

Q6.

(a) [4] What happens when bleaching powder reacts with the following reagents: dil #50) Excess of conc. H2S0 (ii) NH2

(b) [4] How is urea manufactured in Pakistan? Describe in detail.

Q7.

(a) [4] What is orbital hybridization? Explain the geometry of ethyne by sp hybridization

(b) [4] Explain Friedel-Crafts acylation of benzene along with its mechanism.

Q8.

(a) [4] How can ethyne be converted into: (i) Oxalic acid. (i) Acetaldehyde.

(b) [4] Discuss Syl mechanism for nucleophilic substitution reaction of alkyl halide.

Q9.

(a) [4] Discuss the oxidation of ketones and aldehydes in detail.

(b) [4] How can you prepare the following from ethanic (i) Ethyl alcohol (ii) Ethane. (ii) Sodium acetate (iv) Acetic anhydride.

GROUP – II

Objective Type

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

> Note: Objective Type

Q1. The anhydride of HClO4 is:

(A) ClO
(B) Cl2O
(C) Cl2O
(D) Clo2

Q2. The electrophile in aromatic sulphonation is:

(A) HISO
(B) H2SO
(C) SO
(D) SO

Q3. Alkane nitriles can be converted into carboxylic acids by:

(A) Hydration
(B) Oxidation
(C) Hydrogenstion
(D) Acid hydrolysis

Q4. The normal amount of overhead ozone is about:

(A) 150 DU
(B) 250 DU
(C) 350 DU
(D) 450 DU

Q5. Which statement is incorrect:

(A) All the metals are good conductor of electricity
(B) All the metals are good conductor of heat
(C) All the metals form scid oxides
(D) Acid as well as a base

Q6. Which of the following element is present in all amino acids:

(A) HO
(B) C
(C) Br
(D) [illegible in source scan]

Q7. Ethers show the phenomenon of:

(A) Position isomerism
(B) Cis-trans isomerisan
(C) Functional group isomerism
(D) [illegible in source scan]

Q8. Which one of the following does not belong to alkaline earth metals

(A) Ra
(B) Rn
(C) Cu
(D) Ba

Q9. Synthetic rubber is made by the polymerization of:

(A) Acetylene
(B) Chloroprene
(C) Divinyl acetylene
(D) Metamerism

Q10. Which one of the following is a typical transition element:

(A) Sc
(B) Zn
(C) Y
(D) Co

Q11. Elimination bimolecular reactions involve

(A) First order kinetics
(B) Zero order kinetics
(C) Second order kinetics
(D) Third order kinetics

Q12. Which catalyst is used in the contact process:

(A) Fe
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q13. Cannizzaro's reaction is given by:

(A) Formaldehyde
(B) 3-pentanone
(C) Acetaldehyde
(D) Acetone

Q14. Which one is argillaceous material:

(A) Clay
(B) Marble
(C) Lime
(D) Marine shell

Q15. According to Lewis concept ethers behave as

(A) Acid
(B) Base
(C) Acid as well as a base
(D) None of these

Q16. Aluminium oxide is:

(A) Acidic oxide
(B) Basic oxide
(C) Amphoteric oxide
(D) None of these

Q17. Which compound is called a universal solvent:

(A) CH_3OH
(B) $CHOH$
(C) CH_3-O-CH_3
(D) HO

Essay Type

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) How does orthoboric acid react with: (a) NaOH (b) CHOH

(ii) Why are aluminium sheets said to be free from corrosion?

(iii) What is water glass? How is it prepared? Give reaction.

(iv) How is ethyl benzene obtained from bromobenzene? Give reaction

(v) Give structures of the monomers of polyvinyl chloride and polystyrene.

(vi) What is the difference between oligosaccharides and polysaccharides?

(vii) How is quality of raw water improved by aeration process?

(viii) Why is ozone layer depleting?

(ix) How is plastic recycled by depolymerization?

(x) What is meant by addition polymerization? Give an example.

(xi) Why is nitro group meta-directing?

(xii) How is maleic acid obtained from benzene? Give reaction.

Q3. (Marks 16) Write short answers to any EIGHT (8) questions:

(i) What is catalytic cracking? Give its importance.

(ii) What are alicyclic compounds? Give example.

(iii) Convert ethyne into acrylonitrile.

(iv) What are epoxides? How these are prepared from alkene?

(v) Give nitration reaction of methane.

(vi) How does conc. H2S0, reacts with "C and S"?

(vii) Give the effect of heat on H3PO4

(viii) What is laughing gas? How is it prepared?

(ix) Define leaving group. Give some examples.

(x) Convert CH,Mgle into ethane.

(xi) What are fertilizers? Why they are needed?

(xii) Mention raw materials used for cement.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

(i) Define the terms: (a) Coordination sphere. (b) Ligand.

(ii) Give the systematic names to following complexes: (a) [Fe(20% (b) Nay[CF]

(iii) What is sacrificial corrosion?

(iv) How will you distinguish between alcohol and phenol?

(v) How will you convert methanol into ethanol?

(vi) Why ethanol has higher boiling point than diethyl ether?

(vii) How are ketones oxidized by KCO2 / H2SO2?

(viii) Give the mechanism of addition of HCN to acetone.

(ix) What is the strecker's synthesis?

> Note: SECTION-II

> Note: Note: Attempt any THREE questions.

Q5.

(a) [4] Describe the metallic and non-metallic character of elements in the modern periodic table.

(b) [4] How is sodium metal prepared by Down's cell? Explain it.

Q6.

(a) [4] Give any four applications of noble gases.

(b) [4] What is setting of cement? Give the reaction taking place in 1" 24 hours.

Q7.

(a) [4] What is functional group? Give three examples of oxygen containing functional groups.

(b) [4] Explain Friedel-Crafts alkylation of benzene alongwith its mechanism.

Q8.

(a) [4] What is polymerization? Describe the linear and cyclic polymerization of alkynes.

(b) [4] By using Grignard reagent prepare: (i) Primary alcohol. (ii) Carboxylic acids. 9.

Q9.

(a) [4] Describe mechanism of reaction of aldehyde (having no a-hydrogen) in which both oxidation and reduction takes place simultaneously.

(b) [4] Write the mechanism of ester formation by using carboxylic acid and ethanol.